ECO Management Platform (ECOMP)

Product Requirements Document (PRD)

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1. Purpose

The ECO Management Platform (ECOMP) is a **standalone, customer-facing workflow and collaboration portal** designed to manage the full lifecycle of Engineering Change Orders (ECOs). The system replaces spreadsheet- and email-driven processes with a **configurable, auditable, role-based workflow** that supports internal execution by the **Contract Manufacturer (CM)** and secure participation by the **Customer** (approvals, document uploads, visibility).

This PRD defines **what must be built**, not how it will be built.

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2. Goals & Success Criteria

2.1 Primary Goals

- Provide a **single system of record** for ECO execution
- Support **Customer participation** without oversharing internal detail
- Enable **per-ECO customization** of workflow steps
- Improve **cycle time, accountability, and visibility**
- Meet **enterprise security, audit, and compliance expectations**

2.2 Success Metrics

- Reduction in ECO cycle time and wait states
- Reduction in ad-hoc status emails and manual trackers
- % of ECOs with complete documentation at first Customer submission
- SLA adherence by phase and role
- Customer adoption and approval turnaround time

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3. Users & Personas

Internal Users (CM)

- **TPM (Technical Project Manager)** – owns ECO setup, coordination, execution
- **QS (Quote Specialist)** – sourcing, pricing, vendor coordination

- **Materials** – supplier engagement and agreements
- **QE (Quality Engineer)** – FAI, PSW, compliance deliverables
- **Engineering / Production** – FAT/FAI execution, ABI updates
- **Leadership / Executives** – portfolio-level visibility and accountability

External Users (Customer)

- **Customer Admin** – manages Customer users and approvers
- **Customer Approver** – approves gated steps
- **Customer Contributor** – uploads documents, responds to questions
- **Customer Viewer** – read-only status visibility

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4. Core Design Principles

1. **Workflow as Data (not hard-coded BPMN)**
2. **Roles own steps; people are assigned at ECO setup**
3. **Customer visibility is a projection, not the full internal plan**
4. **Security and auditability are first-class requirements**
5. **Modern, intuitive UI comparable to Jira / Monday.com**

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5. Workflow Model

5.1 Global Template

The system must support one or more **Global ECO Templates** representing the complete end-to-end internal ECO process used by the CM.

A Global Template includes:

- Milestones
- **Steps and sub-steps (hierarchical)**
- Default dependencies
- Default role ownership
- Default Customer visibility flags
- Conditional gates (Yes/No decision points)

5.2 ECO Instance Customization

When creating a new ECO from a template, authorized users must be able to:

- Add, remove, or disable steps **and sub-steps**
- Drag-and-drop reorder steps and sub-steps
- Modify dependencies

- Configure which steps are Customer-visible or Customer-actionable
- Assign **roles**, then map **individuals** to those roles for this ECO

All changes must be fully auditable.

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6. Workflow Execution Engine

The system executes workflows using a **state-driven task graph**:

- Tasks move through states (Not Started → In Progress → Blocked → Pending Approval → Done)
- Dependencies control when tasks become available
- Conditional gates enable or disable downstream tasks
- Sequential and parallel execution are supported

SLA clocks must distinguish:

- Active work time
- Waiting on Customer
- Waiting on supplier

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7. Customer Projection Layer (Critical)

The system must maintain **two representations** of an ECO:

1. **Internal Plan** – full detailed workflow (steps + sub-steps)
2. **Customer Projection** – simplified milestone-level view

Customer Projection Requirements

- Aggregates multiple internal steps/sub-steps into Customer-friendly milestones
- Exposes only approved steps, documents, and comments
- Clearly indicates when action is required from the Customer
- Never exposes internal notes, internal-only tasks, or internal SLA data

This projection is configurable per template and per ECO.

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8. Views & User Experience

8.1 Gantt View

- Timeline visualization of milestones, steps, and sub-steps
- Highlights aging, wait states, and critical path blockers

- Supports zoom and collapse/expand

8.2 Kanban View

- Columns by status or milestone
- Swimlanes by role, ECO, or business unit

8.3 Card / List View

- ECO cards showing:
  - Current milestone
  - % complete
  - Days in stage
  - Top blocker
  - Next required action

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9. Approvals & Decision Routing

The system must support:

- Single-approver approvals
- Sequential approvals
- Parallel approvals
- Internal and Customer approvals
- Rejection with required rework loops

Approval actions must:

- Be authenticated
- Be auditable
- Trigger downstream workflow changes

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10. Document Management

Requirements

- Drag-and-drop uploads
- Step- and sub-step-level document requirements
- Versioning and history
- Required document validation
- Customer upload support
- Read/write permissions by role

11. Authentication & Security (Standalone Portal)

11.1 Authentication Model

The platform uses **Customer corporate SSO (federated identity)** as the primary authentication mechanism.

Supported protocols:

- **SAML 2.0**
- **OpenID Connect (OIDC)**

Passwords are **not stored** by the platform for SSO users.

11.2 Identity vs Authorization

- Customer Identity Provider supplies **identity only**
- The platform owns **authorization and permissions**
- Roles are assigned within the platform and **never trusted from IdP claims**

11.3 Tenant Isolation

- Each Customer is a strict tenant
- No cross-tenant data visibility
- Users can only access ECOs belonging to their tenant

11.4 Security Controls

- MFA enforced via Customer IdP
- Signed and validated tokens
- Short-lived sessions
- Re-authentication for sensitive actions
- Encryption in transit and at rest
- Full audit logging

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12. Role-Based Access Control (RBAC)

RBAC is enforced at the API level:

- Role determines allowed actions
- Assignment determines scope
- Visibility flags determine what is shown

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13. Dashboards & Reporting

Worker Dashboard

- My tasks
- Tasks awaiting my approval
- Overdue items
- Blocked items

Executive Dashboard

- ECOs by stage, business unit, Customer
- Aging by milestone
- Wait-state attribution
- SLA adherence

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14. Audit & Compliance

The system must provide:

- Immutable audit logs
- Traceability of all actions
- Approval and document history
- Exportable audit records

Designed to support SOC-2-aligned controls.

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15. Out of Scope (Initial Release)

- Direct modification of ERP / PLM systems
- CAD or BOM authoring
- Full analytics data warehouse

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16. Key Risks & Mitigations

Risk	Mitigation
Oversharing with Customers	Projection layer + visibility flags
Workflow complexity	Template discipline + step/sub-step hierarchy
Long wait states	SLA tracking + attribution
Security exposure	SSO + RBAC + audit

17. Summary

ECOMP delivers a **secure, enterprise-grade, Customer-collaborative ECO platform** that supports complex internal execution by the CM while presenting a controlled, milestone-level experience to Customers. The design intentionally balances flexibility, transparency, and control to enable scalable ECO execution.

Layer A – Architectural Invariants (Mandatory)

18. Template & Instance Governance

18.1 Template Versioning

- Templates are **versioned**.
- A **published template version is immutable**; edits create a new version.
- ECO instances are created from and bound to a **specific template version**.

18.2 In-Flight Instance Changes

- Any modification to steps, sub-steps, dependencies, visibility, gating, or role ownership creates an auditable **Plan Change Event**.
- The system must provide a **diff view** (before/after) including who changed what and when.

18.3 Validation Rules

- No orphaned tasks.
- Dependency integrity must be enforced.
- Circular dependencies must be blocked unless explicitly supported.
- Projection mappings must be revalidated after plan changes.

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19. Customer Projection Mapping Rules

19.1 Mapping Model

- Each Customer milestone is a **projection node**.
- Each projection node maps to one or more internal steps/sub-steps.

19.2 Roll-Up Semantics

- A projection node is **Done** only when all mapped internal tasks are Done.
- A projection node is **Waiting on Customer** if any mapped task is waiting on Customer.

- A projection node is **Blocked** if any mapped task is blocked.
- Aging reflects time in the projection milestone only and must not expose internal SLA thresholds.

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20. Approval Semantics

- Approval decisions are immutable audit events.
- Changes to required documents after approval must trigger configurable re-approval behavior.
- Customer Admins may reassign approvers; all changes must be auditable.
- Rejections must capture a reason and reopen upstream work.

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21. Tenant & SSO Onboarding

- Each Customer tenant must support independent IdP configuration (SAML or OIDC).
- The platform must store a stable external identity identifier (not email).
- Login routing must support tenant-specific entry points and/or email discovery.
- Authorization is always owned by the platform.

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22. Non-Functional Requirements (NFRs)

- Backup and restore must be supported for ECOs, documents, and audit logs.
- The system must scale to **large workflows** and **large document packages**.
- Observability must include logs, metrics, and alerting hooks.
- Audit logs must be exportable.